Caifeng Zou

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website

Seismological Laboratory
California Institute of Technology
1200 E. California Blvd., Pasadena, CA 91125

EDUCATION

California Institute of Technology 2022 –
Ph.D. in Geophysics (Minor in Computational Science and Engineering)
Advisors: Zachary E. Ross & Robert W. Clayton

California Institute of Technology 2022 – 2024
M.S. in Geophysics

Imperial College London 2021 – 2022
M.Sc. with Distinction in Applied Computational Science and Engineering

Tongji University 2016 – 2020
B.S. in Geophysics
GPA: 94.32%

PUBLICATIONS

1. **Zou, C.**, Shi, Y., Ross, Z. E., Clayton, R. W., & Azizzadenesheli, K. (2026). Enforcing Reciprocity in Operator Learning for Seismic Wave Propagation. *Seismological Research Letters*.

2. Kong, Q., **Zou, C.**, Choi, Y., Matzel, E. M., Azizzadenesheli, K., Ross, Z. E., Rodgers, A. J., & Clayton, R. W. (2026). Reducing Frequency Bias of Fourier Neural Operators in 3D Seismic Wavefield Simulations through Multistage Training. *Seismological Research Letters, 97*(1), 272-282.

3. **Zou, C.**, Ross, Z. E., Clayton, R. W., Lin, F. C., & Azizzadenesheli, K. (2025). Ambient Noise Full Waveform Inversion with Neural Operators. *Journal of Geophysical Research: Solid Earth, 130*(11), e2025JB031624.

4. **Zou, C.**, & Clayton, R. W. (2025). Imaging the Northern Los Angeles Basins with Autocorrelations. *Seismological Research Letters, 96*(3), 1791-1801.

5. Zheng, H., Chu, W., Zhang, B., Wu, Z., Wang, A., Feng, B., **Zou, C.**, Sun, Y., Kovachki, N. B., Ross, Z. E., Bouman, K., & Yue, Y. (2025). Inversebench: Benchmarking Plug-and-Play Diffusion Models for Scientific Inverse Problems. *In The Thirteenth International Conference on Learning Representations*.

6. **Zou, C.**, Azizzadenesheli, K., Ross, Z. E., & Clayton, R. W. (2024). Deep Neural Helmholtz Operators for 3D Elastic Wave Propagation and Inversion. *Geophysical Journal International, 239*(3), 1469-1484.

7. **Zou, C.**, Zhao, L., Hong, F., Wang, Y., Chen, Y., & Geng, J. (2023). A Comparison of Machine Learning Methods to Predict Porosity in Carbonate Reservoirs from Seismic-Derived Elastic Properties. *Geophysics, 88*(2), B101-B120.

8. **Zou, C.**, Zhao, L., Xu, M., Chen, Y., & Geng, J. (2021). Porosity Prediction with Uncertainty Quantification from Multiple Seismic Attributes Using Random Forest. *Journal of Geophysical Research: Solid Earth, 126*(7), e2021JB021826.

9. Zhao, L., **Zou, C.**, Chen, Y., Shen, W., Wang, Y., Chen, H., & Geng, J. (2021). Fluid and Lithofacies Prediction Based on Integration of Well-Log Data and Seismic Inversion: A Machine-Learning Approach. *Geophysics, 86*(4), M151-M165.

PREPRINT

1. Shi, Y., Lavrentiadis, G., Tsalouchidis, K., Ross, Z. E., McCallen, D., **Zou, C.**, Azizzadenesheli, K., & Asimaki, D. (2026). Large-Scale 3D Ground-Motion Synthesis with Physics-Inspired Latent Operator Flow Matching. arXiv preprint arXiv:2603.17403.

MANUSCRIPTS IN PREPARATION

1. **Zou, C.**, Shi, Y., Ross, Z. E., Muir, J. B., Olivier, G., Higueret, Q., & Smith, N. (2026). Data-Learned Priors for Posterior Sampling in Full Waveform Inversion.
2. **Zou, C.**, Shi, Y., Ross, Z. E., Clayton, R. W., & Lin, F. C. (2026). Sim-to-Real Full Waveform Inversion of the Los Angeles Basin Using Machine Learning.

INVITED TALKS

Seismic Wavefield Modeling and Inversion Using Neural Operators Nov 16, 2026
SIAM Conference on Imaging Science

Ambient Noise Full Waveform Inversion with Neural Operators Feb 5, 2026
UCLA Geophysics Seminar

HONORS AND AWARDS

GPS Graduate Fellowship, Caltech	Sep 2022
Best Student Presentation, SEG Borehole Geophysics Workshop, Beijing	Nov 2020
Outstanding Graduation Thesis of Tongji University	Jun 2020
Outstanding Graduate of Tongji University	Jun 2020
Geophysical Scholarship, Tongji University	Dec 2019
National Scholarship of China	Nov 2019
Second Prize, China Undergraduate Mathematical Contest in Modeling, Shanghai	Dec 2018
National Scholarship of China	Nov 2018
First-Class Scholarship of Tongji University	Dec 2017
Third Prize, Mathematics Competition for College Students, Shanghai	Nov 2017
Second Prize, Mathematics Competition of Tongji University	Jul 2017

PROFESSIONAL SERVICE

Reviewer for
Geophysical Journal International (3 reviews)
Geophysics (10 reviews)
Geophysical Prospecting (2 reviews)
Earth and Space Science (2 reviews)
Journal of Geophysical Research: Machine Learning and Computation (1 review)

Organizer for the monthly Caltech Seismo Lab Coffee Hour on Machine Learning 2026

Co-convener for session 2026
Developing Data-Driven Methods in the AI Era: New Approaches to Earthquake Science
Seismological Society of America Annual Meeting (Pasadena, CA)

Co-convener for session 2025
Advances in Machine Learning for Solid Earth Geoscience
American Geophysical Union Annual Meeting (New Orleans, LA)

Organizer for the weekly Caltech Seismo Lab Seminar 2024 – 2025

TEACHING

Machine Learning in Geophysics	Teaching Assistant	Caltech	FA 2025 – 26
			FA 2024 – 25
Projects in Machine Learning	Teaching Assistant	Caltech	FA 2024 – 25
Seismology	Teaching Assistant	Caltech	SP 2023 – 24